

[5.4]

- Again, let's restate the following definitions

(a) Micro-canonical Ensemble:

- Closed, isolated system

(i) Total energy E constant.

(ii) Total # of particles N constant

(b) Canonical Ensemble:

(i) N fixed

(ii) In contact with a heat bath

So T can vary.

→ A heat bath (or reservoir): System with much more expendable energy: can absorb and release "infinite" amounts of heat/energy without affecting its temperature T_R

ex: an ideal case is the sun. If we "add" heat to it it will not affect its main temp T_{sun} . likewise the release of light/energy, in short periods of time, also do not affect it.

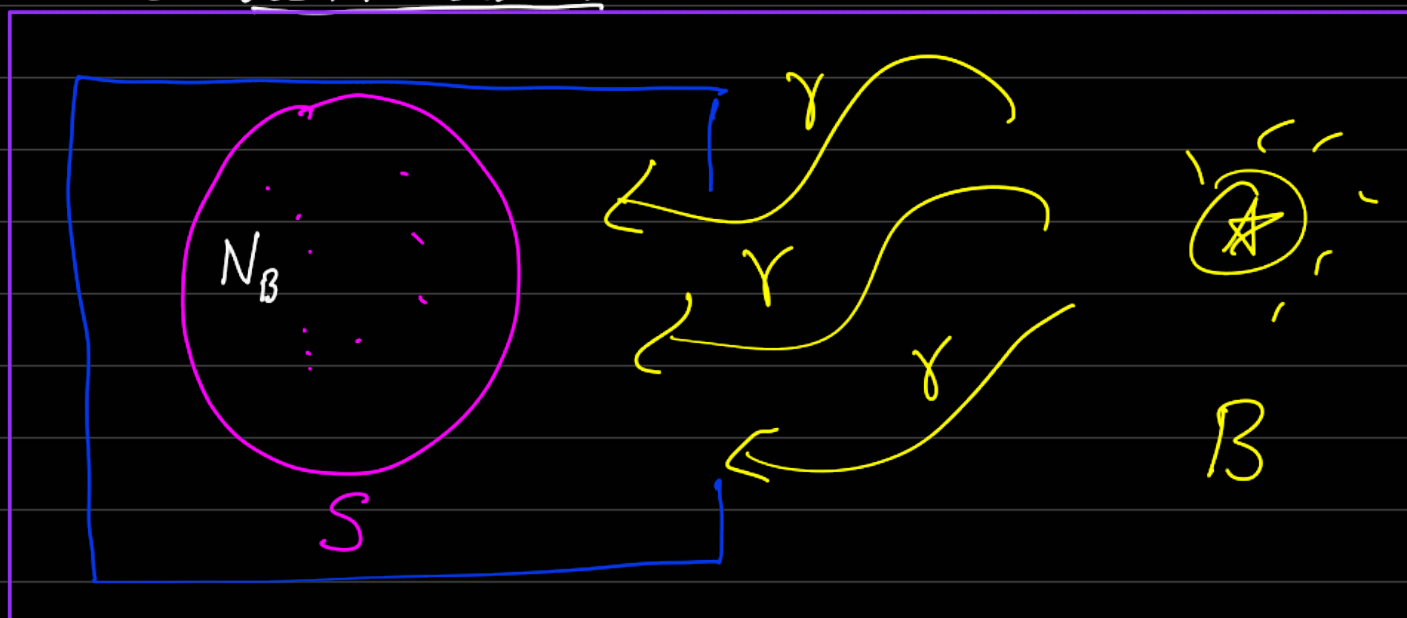
(c) Grand-canonical ensemble:

(i) free to exchange both E & N

* Focus on micro & canonical.

- In my last set of notes we looked at micro-canonical since the systems were isolated.

- Let S be our system and B be the heat bath:



* There's a technicality here!

(i) If we set up our system as the Box $\square$ then the system is a canonical ensemble since the number of particles N_B is the same but our system S is exposed to the heat bath B .

(ii) If we take $\square$ as our system then we have a micro-canonical ensemble since we are keeping all N_B & E in our system.

- We're going to work towards defining the Partition function. and in general it's much easier to calculate this in the canonical ensemble.

Recall: $W(E_B)$: (# of microstates with energy E_B)

Entropy:

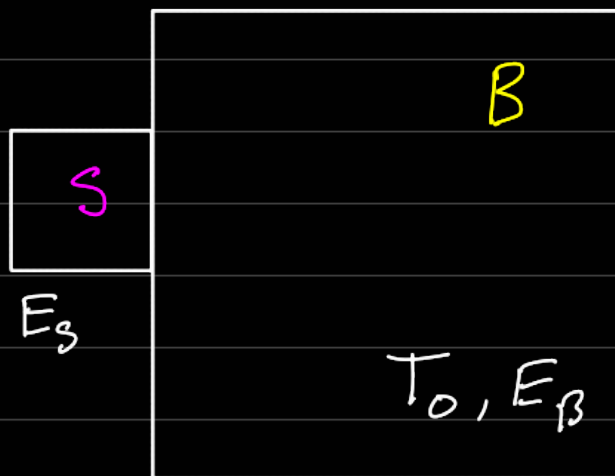
$$S(E) = k_B \ln(W(E))$$

- For a joint system we found

$$W(E_B, E_S) = W(E_B) W(E_S)$$

so

$$\begin{aligned} S(E_B, E_S) &= k_B \ln(W(E_B) W(E_S)) \\ &= k_B \ln(W(E_B)) + k_B \ln(W(E_S)) \\ &= S_B + S_S \end{aligned}$$



* Reservoir is of equilibrium w/ temp T_0

- The total energy of this system is

$$E_{\text{tot}} = E_{B+S} = E_B + E_S$$

- We now want to find the probability of a microstate n of microstate S with energy E_n

(Recall micro states: $\hat{H}|n\rangle = E_n |n\rangle$)

- The Princ. of Equal Prior Prob. states
"An isolated system in equilibrium will have all accessible microstates equiprobable"

Since $(B+S)$ is a micro-canonical system this means

$$P(E_n) = \frac{W_n(E_n)}{W_{\text{tot}}}$$

$$W(E) = \dots$$

$W_n(E_n)$ - Counts the # of microstates of the reservoir **B** associated with the single microstate n of the system **S** for energy E_n .

- In your notes we have:

$$W_n = W(E_n, E_B) = W_S(E_n) \cdot W_B(E_B)$$

Since we have already singled out a specific microstate for **S**

$$W_S(E_n) = 1$$

Thus

$$W_n(E_n) = W_B(E_B = E - E_n)$$

thus

$$P(E_n) = \frac{W_B'(E_B = E - E_n)}{W_{tot}}$$

W_{tot} : total # of microstates for the combined (S+B) - System

* Since the Reservoir **B** is in equilibrium we know

$$\frac{1}{T_0} = \frac{\partial S_B}{\partial E_B} = \frac{\partial (k_B \ln(W_B))}{\partial E_B}$$

$$\frac{1}{T_0} = k_B \frac{\partial \ln(W_B)}{\partial E_B}$$

Solve this:

$$\int \frac{1}{k_B T_0} dE_B = \int d \ln(W_B)$$

$$\frac{1}{k_B T_0} E_B + C = \ln(W_B)$$

$\Rightarrow$

$$W_B(E_B) = \underbrace{e^C}_{\text{constant}} e^{E_B/k_B T_0}$$

Some constant

$$A \equiv e^C$$

$$W_B(E_B) = A e^{E_B/k_B T}$$

But since $E_B = E - E_n$ and we assumed E is constant

$$W_B(E_B) = A e^{(E-E_n)/k_B T_0} = \underbrace{A e^{E/k_B T}}_{\text{Another constant, call it } B} \cdot e^{-E_n/k_B T_0}$$

Another constant, call it B

$$W_B(E_B = E - E_n) = B e^{-E_n/k_B T_0}$$

* Plug this back into our prob.

$$P_n(E_n) = \frac{W_B(E_B = E - E_n)}{W_{\text{tot}}} = \frac{B e^{-E_n/k_B T}}{W_{\text{tot}}}$$

Almost there!

Now, we know that to find $W_B(E - E_n)$

we assumed we looked at a single state in S the total must then be the sum over all possible states $|n\rangle$ in S

$$W_{\text{tot}} = \sum_{n=1}^{N_S} W_B(E - E_n)$$

$$= \sum_{n=1}^{N_S} (B e^{-E_n/k_B T_0})$$

$$= B \sum_{n=1}^{N_S} e^{-E_n/k_B T_0}$$

thus,

$$W_{\text{tot}} = B \sum_{n=1}^{N_S} e^{-E_n/k_B T_0}$$

lets plug this in:

$$\cancel{B} e^{-E_n/k_B T}$$

$$P_n(E_n) = \frac{e^{-E_n/k_B T}}{\sum_i e^{-E_i/k_B T}}$$

$$P_n(E_n) = \frac{e^{-E_n/k_B T_0}}{\sum_i e^{-E_i/k_B T_0}}$$

(prob. of finding the system at an energy E_n)

Define: Partition function:

$$Z \equiv \sum_i e^{-E_i/k_B T_0}$$

$$P_n(E_n) = \frac{e^{-E_n/k_B T_0}}{Z}$$

*Note: We've taken the Reservoir into account and in particular T_0 in the eqs above are the temp of **B**. It is the only remaining information of B's influence.

• We see if $\tilde{E}_n \gg k_B T$ then this state has a low prob. of being occupied. This makes sense a state with $\tilde{E}_n$ would on average have Energy greater than those from the heat Bath.

• $T \rightarrow 0$ Forces dist. into the ground state (there's no "excitation" energy)

$T \rightarrow \infty$ We saw, in my last notes that as $T \rightarrow \infty$ we achieve equilibrium.

$$\lim_{T \rightarrow \infty} e^{-E_n/k_B T} = e^0 = 1$$

$$\text{so } \lim_{T \rightarrow \infty} (Z) = \sum_{i=1}^N \lim_{T \rightarrow \infty} (e^{-E_i/k_B T}) = \sum_{i=1}^N 1$$

$$= N$$

$$\text{Thus: } \lim_{t \rightarrow \infty} (p(n)) = \frac{1}{N}$$

↳ This is exactly what we expected.

* Extra: You can skip this: just read last paragraph of this section

Z is often seen as a rather abstract and, well, indeed it is. Here I'll show how Z can be interpreted from the Quantum point of view

$$\beta \equiv \frac{1}{k_B T}$$

• Trace of a quantum operator: $\hat{A}$

$$\text{Tr}(\hat{A}) \equiv \sum_n \langle n | \hat{A} | n \rangle$$

$$\text{Thus, for } \hat{A} = e^{-\beta \hat{H}}$$

where $\hat{H}$ is the Hamiltonian

$$\text{Tr}[e^{-\beta \hat{H}}] = \sum_n \langle n | e^{-\beta \hat{H}} | n \rangle = \sum_n \langle n | e^{-\beta E_n} | n \rangle$$

Since

$$\hat{H} | n \rangle = E_n | n \rangle$$

$$\text{Tr}[e^{-\beta \hat{H}}] = \sum_n e^{-\beta E_n} \underbrace{\langle n | n \rangle}_1$$

Thus:

$$Z = \text{tr}(e^{-\beta \hat{H}}) = \sum_n e^{-\beta E_n}$$

Furthermore,

$$\mathbb{I} = \sum_n | n \rangle \langle n |, \quad \mathbb{I} = \int dq | q \rangle \langle q |$$

$$\mathbb{I} = \int dq | q' \rangle \langle q' |$$

$$\begin{aligned}
 Z &= \sum_n \langle n | \int dq |q\rangle \langle q| e^{-\beta H} \int dq' |q'\rangle \langle q'| n \rangle \\
 &= \int dq dq' \langle q | e^{-\beta H} | q' \rangle \sum_n \underbrace{\langle q' | n \rangle \langle n | q \rangle}_1 \\
 &= \int dq dq' \langle q | e^{-\beta H} | q' \rangle \langle q' | q \rangle
 \end{aligned}$$

but

$$\langle q | q' \rangle = \delta(q - q')$$

so

$$Z = \int dq dq' \langle q | e^{-\beta H} | q' \rangle \delta(q - q')$$

$$Z = \int dq \langle q | e^{-\beta H} | q \rangle$$

We know $\hat{H}(p, q)$: (phase space Hamiltonian)
given by

$$\hat{H}(p, q) = \frac{\hat{p}^2}{2m} + V(\hat{q})$$

• Introduce

$$1 = \int dp |p\rangle \langle p|$$

$$1 = \int dp' |p'\rangle \langle p'|$$

as above

$$Z = \int dq \langle q | \int dp |p\rangle \langle p| e^{-\beta p^2/2m} \int dp' |p'\rangle \langle p'| q \rangle \times e^{-\beta V(q)}$$

$$= \frac{1}{2\pi\hbar} \int dp dq e^{-\beta H(p, q)}$$

Using

$$\langle q | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipq/\hbar}$$

$$Z = \sum_n e^{-E_n/k_B T} = \frac{1}{2\pi\hbar} \int dq dp e^{-\beta H(p, q)}$$

↑

(Quantum)

↑

(Classical)

(Partition f^n)

(Part. f^n)

* For $\hat{H}(q, p) = \hat{p}^2/2m$

$$Z = \frac{1}{2\pi\hbar} \left(\int dq \right) \left(\int dp e^{-\beta p^2/2m} \right)$$

• The term $(\int dp e^{-\beta p^2/2m})$ is a Gaussian int.

$$\int dx e^{-ax^2} = \sqrt{\frac{\pi}{a}}$$

• The term $(\int dq)$ is an integral over all the space: i.e. it is the Volume

$$\int dq \equiv V$$

thus

$$Z = V \left(\frac{m k_B T}{2\pi\hbar^2} \right)^{3/2}$$

thermal de-Broglie wavelength: $\lambda \equiv \sqrt{\frac{2\pi\hbar^2}{m k_B T}}$

$$Z = \frac{V}{\lambda^3}$$

* Thus: Z counts the number of thermal wavelengths that can fit into a Volume V .

It tells us the how many microstates are accessible to your system in some ensemble: Measures the volume occupied by the system in phase space.

[5.5.1] & [5.5.2] (continued)

* We've gone through the effort to define the Part. Function $Z = \sum_i e^{-E_i/k_B T}$

$$P_n(E_n) = \frac{e^{-E_n/k_B T}}{Z} \quad \Bigg| \quad Z = \sum_i e^{-E_i/k_B T}$$

but why?

Well, in my last notes we found via very involved process that for the two state system

$$E_{\uparrow} = E, \quad E_{\downarrow} = 0$$

and

$$N = N_{\uparrow} + N_{\downarrow}$$

$$E_{\text{tot}} = N_{\uparrow} E_{\uparrow} + N_{\downarrow} E_{\downarrow} \\ = N_{\uparrow} E + 0$$

$$E_{\text{tot}} = N_{\uparrow} E$$

• We found: (Pg 13)

$$\frac{N_{\uparrow}}{N} = \frac{E}{N E} = \frac{1}{e^{E/k_B T} + 1}$$

Lets see the power of our new tools:

$$P(E_{\uparrow}) = \frac{e^{-E_{\uparrow}/k_B T}}{Z} = \frac{e^{-E/k_B T}}{Z}$$

$$Z = \sum_i e^{-E_i/k_B T} = e^{-E_{\uparrow}/k_B T} + e^{-E_{\downarrow}/k_B T} \\ = e^{-E/k_B T} + e^0 = e^{-E/k_B T} + 1$$

$$Z = e^{-E/k_B T} + 1$$

plugging this in:

$$P(E_{\uparrow}) = \frac{e^{-E/k_B T}}{e^{-E/k_B T} + 1}$$

• Multiply by 1: $\frac{1}{1 + e^{E/k_B T}} = \frac{e^{-E/k_B T}}{e^{-E/k_B T} + 1}$

$$p(E_{\uparrow}) = \left(\frac{e^{-E_{\uparrow}/k_B T}}{e^{-E_{\uparrow}/k_B T} + 1} \right)$$

so,

$$p(E_{\uparrow}) = \frac{1}{e^{E_{\uparrow}/k_B T} + 1}$$

But if we have N total states and $N_{\uparrow}$ states the macro-state probability is simply

$$p(E_{\uparrow}) = \frac{N_{\uparrow}}{N}$$

Equating these two

$$\frac{N_{\uparrow}}{N} = \frac{1}{e^{E_{\uparrow}/k_B T} + 1}$$

Since $E = N_{\uparrow} E \Rightarrow N_{\uparrow} = \frac{E}{E}$

$$\frac{E}{EN} = \frac{1}{e^{E/k_B T} + 1}$$

as before! Thus, we see our new tools make computation MUCH easier!

